

Comprehensive Report: Text Representation and Unsupervised Learning

[Link to the notebook](#)

Executive Summary

This notebook developed an end-to-end workflow for analyzing text with unsupervised machine learning. The Reuters corpus was loaded and converted into a single-label dataset by assigning each document its first available Reuters category. The text was then normalized through lowercasing, tokenization, stopwords removal, and punctuation filtering. Two sparse numerical representations were created: Bag of Words (BoW) and TF-IDF. K-Means clustering was applied to the TF-IDF matrix, cluster terms were inspected for interpretation, and TruncatedSVD was used to create a two-dimensional visualization suitable for sparse text data. Finally, documents held out from model fitting were assigned to clusters and compared with their original categories.

The workflow shows that TF-IDF is a practical representation for topic-oriented clustering because it reduces the influence of common words and emphasizes terms that distinguish documents. However, the resulting clusters are unsupervised groupings rather than guaranteed reproductions of the Reuters categories, and cluster identifiers do not have an inherent semantic meaning.

1. Dataset and Objective

The Reuters corpus provides news documents organized into topical categories. The analysis had four main goals:

1. Prepare raw documents for numerical analysis.
2. Compare count-based and importance-weighted text representations.
3. Discover document groupings with K-Means clustering.
4. Interpret and assess those groupings using representative terms, a silhouette score, a two-dimensional projection, and predictions for held-out documents.

The Reuters labels were retained as reference information. They were not used as features when fitting K-Means, so the clustering stage remained unsupervised.

2. Text Preparation Methodology

The corpus was loaded with NLTK, including the Reuters documents, category metadata, English stopwords, and the tokenizer. Documents without a category were removed. For documents with multiple Reuters categories, the first category was selected to create one label per document. This

simplifies the demonstration, but it also means that the reference labels do not capture the full multi-label nature of the original corpus.

The preprocessing function applied the following transformations:

- Converted text to lowercase to avoid treating capitalization variants as separate terms.
- Tokenized each document into individual tokens.
- Removed English stopwords that occur frequently but often provide limited topic information.
- Removed punctuation and non-alphanumeric tokens.
- Joined the remaining tokens into normalized text strings.

These steps reduce vocabulary noise and make the later feature matrices more useful for comparing document content.

3. Text Representation

Bag of Words

The Bag-of-Words representation was created with `CountVectorizer`. Each document became a sparse vector whose entries represent raw word counts. This method is easy to interpret and preserves frequency information, but very common words can dominate the representation even when they are not especially useful for distinguishing topics.

TF-IDF

The TF-IDF representation was created with `TfidfVectorizer`. TF-IDF increases the weight of terms that are important within a document while reducing the weight of terms that occur in many documents. This is a suitable representation for clustering news text because topic-specific vocabulary should contribute more strongly than corpus-wide vocabulary.

Both matrices are sparse because most documents contain only a small fraction of the full vocabulary. Preserving sparsity is important for memory efficiency and computational performance.

4. K-Means Clustering Methodology

K-Means was applied to the TF-IDF feature matrix. The number of clusters was set to the number of reference categories, `len(target_names)`, which provides an interpretable starting point for comparing discovered groups with the known Reuters topics. This is a modeling choice rather than a guarantee that the true number of semantic groups equals the number of selected labels.

The model used the following settings:

- `init='k-means++'` to improve the starting centroid locations.
- `n_init=10` to run multiple initializations and select a better solution.
- `max_iter=100` to limit the number of iterations per initialization.
- `random_state=42` to make the fitting process reproducible.

The fitted model produced a cluster assignment for every document and a centroid for every cluster. Because K-Means optimizes distance to centroids, it groups documents according to their TF-IDF geometry rather than directly optimizing agreement with Reuters labels.

5. Results and Evaluation

The notebook reports the number of documents, the number of categories, the feature-matrix dimensions, sample cluster assignments, and the silhouette score. The silhouette score measures how similar each document is to its own cluster compared with neighboring clusters. A higher score indicates stronger cohesion and separation; a score near zero indicates substantial overlap; and a negative value suggests that some documents may fit better in another cluster.

The score is calculated on a sample of up to 2,000 documents to keep evaluation practical. It should therefore be interpreted as an estimate of clustering quality for the corpus, not as an exhaustive evaluation of every document.

The top-term analysis ranks the words with the largest TF-IDF centroid values for each cluster. These terms provide the main qualitative evidence for interpreting the discovered topics. A coherent cluster should contain related terms, while a cluster with broad or mixed vocabulary may represent several related topics or documents that are difficult to separate using the available representation.

6. Dimensionality Reduction and Visualization

The notebook uses `TruncatedSVD` rather than dense PCA for the TF-IDF matrix. `TruncatedSVD` is appropriate for sparse text data because it can reduce dimensionality without converting a potentially large vocabulary matrix into a dense array. The first two components were plotted with points colored by K-Means cluster assignment.

This visualization provides a simplified view of the high-dimensional document space. Nearby points may have similar term usage, and separated regions may suggest broad topical differences. However, two components cannot preserve all information from the original feature space. Overlap in the plot does not necessarily mean the full-dimensional clusters are identical, and apparent separation should be confirmed with quantitative metrics and term inspection.

7. Prediction on Held-Out Documents

A train-test split was used to simulate new documents. K-Means was retrained on the training portion of the TF-IDF matrix, and three documents from the test portion were assigned to their nearest learned centroid. For each document, the notebook prints a shortened processed text, its original Reuters category, its predicted cluster, and representative terms from that cluster.

This demonstrates how the fitted model can assign unseen documents to existing groups. It is not a conventional classification accuracy test because cluster numbers are arbitrary and are not directly equivalent to category names. To compare cluster assignments with categories formally, a mapping from clusters to labels or metrics such as adjusted Rand index, normalized mutual information, or a contingency table would be needed.

8. Key Insights

- Text preprocessing substantially reduces irrelevant variation and creates a more consistent vocabulary.
- BoW is straightforward and interpretable, while TF-IDF is generally better suited to distinguishing document topics.
- Sparse matrix representations are essential when the vocabulary contains many thousands of terms.
- K-Means can reveal broad topical structure, but similar news topics may produce overlapping clusters.
 - The most representative centroid terms are useful for assigning human-readable themes to otherwise arbitrary cluster IDs.
 - The silhouette score provides a compact numerical assessment, but it should be interpreted alongside qualitative term analysis and the projection plot.
 - TruncatedSVD offers an efficient sparse-safe visualization method; the two-dimensional plot is exploratory rather than definitive.
 - Held-out prediction confirms that the model can assign new documents to learned clusters, but it does not by itself establish supervised category accuracy.

9. Limitations and Improvements

The use of the first Reuters category for multi-category documents discards some label information. The chosen number of clusters is based on the number of reference categories rather than a systematic search over candidate values. K-Means also favors compact, centroid-based groups and may not model irregular topic structure well.

The analysis could be strengthened by comparing several values of ``k`` using silhouette score and other internal metrics, evaluating agreement with the reference labels using adjusted Rand index or normalized mutual information, and inspecting cluster size balance. Additional preprocessing could include stemming or lemmatization, while vectorization could be improved with ``ngram_range``, ``min_df``, and ``max_df`` settings. Alternative models such as hierarchical clustering, spherical K-Means, or topic models could provide different views of the corpus.

The random word embeddings in the notebook are conceptual only. They are not learned from the Reuters corpus and should not be interpreted as meaningful semantic embeddings. A production system would use trained embeddings or a validated pretrained language model and would evaluate it against a clearly defined downstream task.

10. Reproducibility and Conclusion

The main clustering and dimensionality-reduction operations use ``random_state=42``, making their results reproducible when the same dataset, library versions, and preprocessing environment are used. NLTK resource downloads and the Reuters corpus must be available before running the analysis. Exact matrix dimensions, top terms, cluster assignments, and the silhouette score should be taken from the executed notebook outputs because they depend on the installed versions and corpus state.

Overall, the notebook demonstrates a complete text-mining pipeline: raw documents are cleaned, converted into numerical features, grouped without using labels, visualized in reduced dimensions, interpreted through representative terms, and applied to held-out documents. The strongest conclusion is that TF-IDF plus K-Means provides a useful baseline for discovering broad themes in Reuters text, while careful evaluation is necessary before treating the discovered clusters as definitive topic categories.